How much of a solar forecast is skill?

A two-site day-ahead study with public data

Saeed Peyman
Independent researcher, Nashville, Tennessee, United States
saeeeed@duck.com

Abstract. Day-ahead forecasts of photovoltaic (PV) output are usually reported as a single error number for one plant. This study asks how much of that number is skill, how much depends on the place, and where the skill runs out. It uses two public plants from the NREL PVDAQ archive in contrasting climates, an 893 kW fixed-tilt array in Arbuckle, California and a 4,738 kW single-axis tracker in Kersey, Colorado, and weather forecasts as they were issued a day ahead (GFS and ECMWF IFS, archived by Open-Meteo), with ERA5 reanalysis as a hindsight comparison. Every forecaster is scored on the same daytime hours with a monthly rolling origin: 6,009 hours over 549 days at Arbuckle and 3,078 hours over 298 days at Kersey. Against smart persistence (normalised RMSE 0.194 and 0.232 of AC rating), gradient-boosted trees on GFS inputs reach a skill of +0.25 (95% block-bootstrap interval +0.20 to +0.29) at Arbuckle and +0.22 (+0.14 to +0.29) at Kersey. A physical model with a two-parameter calibration gets most of the way there (+0.19 and +0.21); at Kersey the trees add nothing measurable over it. Hindsight weather from the reanalysis adds little at Arbuckle (+0.26) and somewhat more at Kersey (+0.29). A model trained on the other site keeps less than half of the gain at Arbuckle (+0.10) and about three quarters at Kersey (+0.16). Nominal 80% intervals cover 59% and 65% of hours. The worst tenth of days carries about a quarter of all absolute error. Read one by one, many of the largest misses are not errors of the weather forecast at all: a plant that produced nothing on two clear days and an inverter offline for a week; a summer of clear, dry days at reduced output may be equipment, soiling or smoke. Days on which the forecast was wrong and the reanalysis right are a minority of the large misses. All data are public, all code is released, and the whole study reruns with one command.

Keywords: solar forecasting; photovoltaics; day-ahead forecasting; numerical weather prediction; gradient boosting; forecast verification; reproducibility.

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Code and data pipeline: https://github.com/speyyman/solar-two-sites (MIT licence). **This text and its figures:** CC BY 4.0.

1 Introduction

A day-ahead PV forecast is a statement made at the end of one day about every hour of the next. Its value to a grid operator or a plant owner depends on how much better it is than what they could have guessed for free. Published forecasting studies report such comparisons, but often for a single plant, with weather inputs whose provenance is not always clear, and with error figures that combine three different things: how well the weather was forecast, how well weather was turned into power, and how often the plant did something no weather model could know about.

This paper separates those three things as far as public data allow. It asks:

1. How much skill do learned models add over honest physical and persistence baselines when they are given only what a forecaster would have had at issue time?

2. How does that skill differ between two contrasting climates, and how much of a model trained at one site carries over to the other?

3. Under what conditions does the forecast break down, and how many of the large misses are about the weather at all?

The study makes three choices that shape every result. First, the weather inputs are true day-ahead forecasts: values that each model issued at least 24 hours before the hour they describe, retrieved from an archive of past runs, not reanalysis or short-lead forecasts stitched together after the fact. Second, the measured power is cleaned by stated rules, and every value removed, shifted or filled in is counted in a ledger that the paper reports (Appendix A). Third, the baselines come first: a model is only credited with what it adds over smart persistence and over a physical model built from the plant's own metadata.

This is a new, independent study. It was prompted by the author's earlier team capstone on solar yield for a site in Jordan, but it shares no code, data or analysis with that project.

2 Related work

Reviews of PV power forecasting [2] describe the usual chain for day-ahead horizons: numerical weather prediction (NWP) for irradiance and temperature, a physical or statistical model from weather to power, and often a statistical correction on top. They also note how hard results are to compare across studies, because sites, horizons, normalisations and baselines differ. Yang et al. [18] set out a verification framework for deterministic solar forecasts built on the skill score against a reference forecast; as the reference they recommend the optimal convex combination of climatology and persistence. This paper uses the simpler persistence of the clear-sky index ("smart persistence") as its reference, so its skill scores are not directly comparable with scores against that stricter reference. For probabilistic forecasts, the pinball loss of quantile regression [14] is a proper scoring rule [7] and was the main metric of the solar track of the Global Energy Forecasting Competition 2014 [11].

On evaluation, rolling-origin out-of-sample testing [17] avoids the optimism of a single split and of random cross-validation on autocorrelated data. Daily errors are themselves autocorrelated over runs of similar weather, so confidence intervals here resample whole blocks of days [15].

The physical chain uses pvlib python [1, 10]: the Ineichen–Perez clear-sky model [12], Hay–Davies transposition [8], the Faiman cell temperature model [6], and the PVWatts DC, loss and inverter models [5]. The learned models are gradient-boosted decision trees from LightGBM [13]. Weather comes from NOAA's Global Forecast System [16], the ECMWF Integrated Forecasting System open data, and the ERA5 reanalysis [9], all through the Open-Meteo API [19]. Plant data come from the PVDAQ archive on the Open Energy Data Initiative [4].

3 Data

3.1 Two plants

The study needs measured output from plants whose records overlap the period for which day-ahead forecasts were archived (Section 3.2). Of the public PVDAQ systems surveyed (the 2023 Solar Data Prize systems and the long-running public systems), two had usable plant-level output in 2024 and 2025 and documented metadata; the survey was not exhaustive:

- **Arbuckle, California** (PVDAQ system 2107). A fixed ground-mounted array on the floor of the Sacramento Valley, hot-summer Mediterranean climate (Köppen Csa). 893 kW DC of monocrystalline modules at 25° tilt facing south, 24 string inverters. Revenue-grade AC meter, 15-minute samples, January 2024 to December 2025.

- **Kersey, Colorado** (PVDAQ system 9068). A single-axis tracker on the high plains east of the Front Range, cold semi-arid climate (BSk), at 1,407 m. 4,738 kW DC of thin-film CdTe modules on two central inverters. January 2024 to April 2025, so 16 months rather than two years.

Errors are normalised by the nominal AC rating: 662.4 kW at Arbuckle (24 inverters of 27.6 kW nominal, 30 kW maximum, per the manufacturer's datasheet) and 3,820 kW at Kersey (two inverters of 1,910 kW, per the PVDAQ metadata). The physical model clips at the inverters' maximum output.

Kersey's revenue meter is not usable in this period: it reports values several times the plant's rating in August and September 2024 and values near zero from October to December 2024, while both inverters report normal production. Kersey's output is therefore the sum of its two inverters' AC power.

Both plants also report plane-of-array irradiance from on-site sensors. These were checked and not used for any result: the Arbuckle 2025 irradiance file is labelled UTC but follows a fixed Pacific standard-time clock for most of the year, and the Kersey sensor implies a performance ratio above one in August 2024.

3.2 Weather inputs

The forecast is issued at the end of local day D−1 for every hour of day D. Its weather inputs must therefore be values that an NWP model had published at least a day earlier. Open-Meteo's Previous Runs API serves, for each hour, the value issued 24 hours before it (the `previous_day1` series). This archive starts in January 2024, so the study window begins on 20 January 2024. Open-Meteo's separate Historical Forecast API is not used: it assembles each hour from the start of a recent run, which describes a shorter lead time than a day-ahead forecaster would have.

Three weather sources are used for each site, hourly, at the plant's coordinates:

- **GFS** (`gfs_seamless`, NOAA's GFS blended with HRRR over the United States), day-ahead. This is the primary input.
- **ECMWF IFS** at 0.25° (open data), day-ahead. A robustness check.
- **ERA5** reanalysis. Not a forecast: it is the best later estimate of the weather that actually happened, used to measure how much of the remaining error is the weather forecast's.

Each provides global, direct normal and diffuse irradiance, cloud cover, 2 m temperature and humidity, 10 m wind speed and precipitation. Irradiance values are means over the preceding hour, so every series in the study is labelled by hour end.

3.3 Cleaning and the ledger

The measured series are cleaned by fixed rules, applied in order: timestamps are read in the clock the file was written in and converted to UTC, dropping the hour that daylight saving makes ambiguous; each file's clock is checked against the sun (the midpoint of production should sit at solar noon) and shifted by whole sampling steps where it does not, and any month whose clock still disagrees with the sun by more than 0.75 h is dropped; negative night-time readings become zero; readings above 105% of the inverters' maximum and runs of an identical non-zero value lasting an hour or more are dropped; the 15-minute samples are averaged to hours, and an hour with fewer than 75% of its samples is left missing. Target values are never filled in.

The Arbuckle meter labels each interval by its start; the clock check found this (a median offset of a quarter of an hour) and the series was shifted by 15 minutes. At Kersey, 1,561 negative night samples were set to zero and 52 hours had too few samples; at Arbuckle, 153 samples were dropped as frozen values and 32 hours had too few samples. The full ledger is in Appendix A.

4 Method

4.1 Forecasters

All forecasters predict hourly AC power for day D, as a fraction of AC rating, from information available at the end of D−1. Night hours are zero and every forecast is clipped to the inverters' range.

Persistence. The measured power at the same local hour on D−1 (or the most recent earlier day with a value, up to seven days back; every such fill is counted in the ledger).

Smart persistence. The ratio of measured to clear-sky energy over the whole of D−1, times the clear-sky power for each hour of D. Clear-sky power is the physical model below driven by Ineichen–Perez clear-sky irradiance at 20°C and 1 m/s wind. This is the reference for every skill score.

Physical. Sun position at mid-hour, Hay–Davies transposition onto the fixed plane or the backtracking tracker (60° limit, ground coverage ratio 0.4), Fresnel incidence-angle loss on beam irradiance, Faiman cell temperature, PVWatts DC power with the module temperature coefficient and the default PVWatts system losses, and the PVWatts inverter model. Nothing is fitted to data; the inputs are the day-ahead forecast, the plant metadata, and three stated assumptions where the metadata are silent: a 60° tracker limit, a ground coverage ratio of 0.4, and at Arbuckle a typical monocrystalline temperature coefficient of −0.37%/°C (Kersey's −0.28%/°C is from its metadata).

Physical, calibrated. A least-squares scale and offset on the physical forecast, refitted on all earlier data at each origin. This two-parameter model is the linear sanity check.

Boosted trees. LightGBM regression on 18 features: the physical and clear-sky power, the forecast's irradiance components, clearness index, cloud cover, temperature, humidity, wind and precipitation, frozen precipitation over the previous 72 hours, solar zenith and azimuth, solar hour (as sine and cosine), and the two persistence values. Hyperparameters were fixed before any test month was scored (400 trees, learning rate 0.05, 31 leaves, row and column subsampling 0.8, L2 penalty 1) and not tuned. Day of year is deliberately not a feature, so that a one- or two-year record cannot memorise last year's weather.

Quantile trees. The same model with the pinball objective for the 10th, 50th and 90th percentiles, sorted per hour so the bands never cross.

Transfer. The boosted trees trained only on the other site's data (to the same cut-off date), then applied with this site's inputs.

4.2 Evaluation

Evaluation is rolling-origin by calendar month. For each test month from July 2024, every learned forecaster is trained on all days before that month and predicts every hour of it; the months before July 2024 (about five and a half months) are training only. Persistence and the uncalibrated physical model need no training. A test that rebuilds the features and checks that every persistence value comes from a strictly earlier day runs on every table the pipeline builds, and a synthetic end-to-end test confirms that a planted leak is caught.

All forecasters are scored on the same hours: daytime (solar zenith below 85°), measured power present, and every forecast present. Metrics are normalised mean absolute error (nMAE), normalised root mean squared error (nRMSE), mean bias, and the skill score

$$S = 1 - \frac{\mathrm{RMSE(forecast)}}{\mathrm{RMSE(smart\ persistence)}},$$

so $S = 0$ is no better than the reference and $S = 1$ is perfect. Intervals for skill scores are 95% percentile intervals from 1,000 bootstrap resamples of whole seven-day blocks. Quantile forecasts are scored by mean pinball loss and by the coverage of the 10–90% band.

Results are broken down by sky condition, using the day's measured energy over clear-sky energy (k): clear ($k \geq 0.8$), mixed ($0.4 \leq k < 0.8$) and overcast ($k < 0.4$). This uses the measured outcome, so it describes where errors fall, not something a forecaster could know in advance.

4.3 Failure analysis

For every scored day, the daily energy error of the primary model (boosted trees on GFS) is computed in units of AC-rated hours (one unit is the energy the plant makes in one hour at its AC rating), together with three daily clearness indices on the same clear-sky base: what the plant saw (k_{obs}, measured over clear-sky energy), what the forecaster was told (k_{nwp}, forecast over clear-sky irradiance) and a later independent estimate of the sky (k_{era5}). Days with an error above one rated hour are sorted by fixed rules using only these indices, ERA5 humidity, wind and frozen precipitation, and, at Kersey, the per-inverter energy. The rules are listed in `pipeline/failures.py`; the thresholds are round numbers, not tuned.

The rules were revised twice, and the revisions are disclosed here because each was made after looking at cases. The first version, written before the cases were read and not preserved, had a single "plant" category for every day on which output fell far below a sky that ERA5 called sunny; reading those cases showed two different things in it, still winter days and dry clear summer days, so the second version split them. The second version also had a "reactive-power limit" category for Kersey days with a low power factor and low output under a sunny reanalysis sky. An independent review of a draft showed that rule was confounded (Section 6), and the third version, used here, drops it. No forecast or headline score changed; the plant-anomaly sensitivity check did.

A separate, forecast-free rule marks "plant-anomaly" days, when the plant rather than the sky most likely limited output: near-zero output under a sunny reanalysis sky, output far below a clear and dry reanalysis sky, or one Kersey inverter producing under a tenth of the other's energy. The headline scores are then recomputed without those days as a sensitivity check.

5 Results

5.1 Headline scores

Table 1 and Figure 1 give every forecaster's scores over the whole test period at each site. Plain persistence is worse than smart persistence at both sites (−0.07 and −0.11), as expected, since it carries yesterday's sun angle into today. The physical model driven only by the GFS forecast and the plant metadata beats smart persistence by +0.08 at Arbuckle and +0.11 at Kersey, with intervals that touch zero; it over-forecasts on average (bias +0.071 and +0.050 of rating). A scale and offset fitted on past data removes most of that bias and roughly doubles the skill (+0.19 and +0.21).

Boosted trees on the same GFS inputs reach +0.25 (+0.20 to +0.29) at Arbuckle and +0.22 (+0.14 to +0.29) at Kersey. The question this study set out to answer is how much of that is the learning. Compared directly with the calibrated physical model on the same bootstrap blocks, the difference in RMSE (trees minus calibrated physical, as a fraction of rating) has a 95% interval of −0.018 to −0.005 at Arbuckle, so there the trees are better. At Kersey the interval is −0.008 to +0.004: no difference that this record can detect. Most of the skill, then, comes from a physical model and two fitted numbers; the trees add a modest amount at one site and nothing measurable at the other.

On the months both sites share (July 2024 to April 2025, 304 and 298 days), smart persistence scores 0.204 at Arbuckle and 0.232 at Kersey, and the trees' skill is +0.27 and +0.22. The ordering of the forecasters is the same as over the full period.

	Arbuckle, CA			Kersey, CO		
Forecaster	nMAE	nRMSE	Skill	nMAE	nRMSE	Skill
Persistence	0.120	0.209	−0.07	0.174	0.258	−0.11
Smart persistence	0.122	0.194	–	0.170	0.232	–
Physical (GFS)	0.123	0.179	+0.08	0.138	0.207	+0.11
Physical, calibrated (GFS)	0.108	0.158	+0.19	0.134	0.184	+0.21
Boosted trees (GFS)	0.096	0.147	+0.25	0.133	0.182	+0.22
Boosted trees (ECMWF)	0.092	0.144	+0.26	0.126	0.174	+0.25
Boosted trees (ERA5, hindsight)	0.091	0.144	+0.26	0.116	0.164	+0.29
Boosted trees, other site (GFS)	0.133	0.175	+0.10	0.144	0.196	+0.16

Table 1: Scores over every scored hour (Arbuckle 2024-07-01 to 2025-12-31, 6,009 hours; Kersey 2024-07-01 to 2025-04-29, 3,078 hours). nMAE and nRMSE are fractions of AC rating; skill is against smart persistence. The ERA5 row uses the weather that happened and is not a forecast.

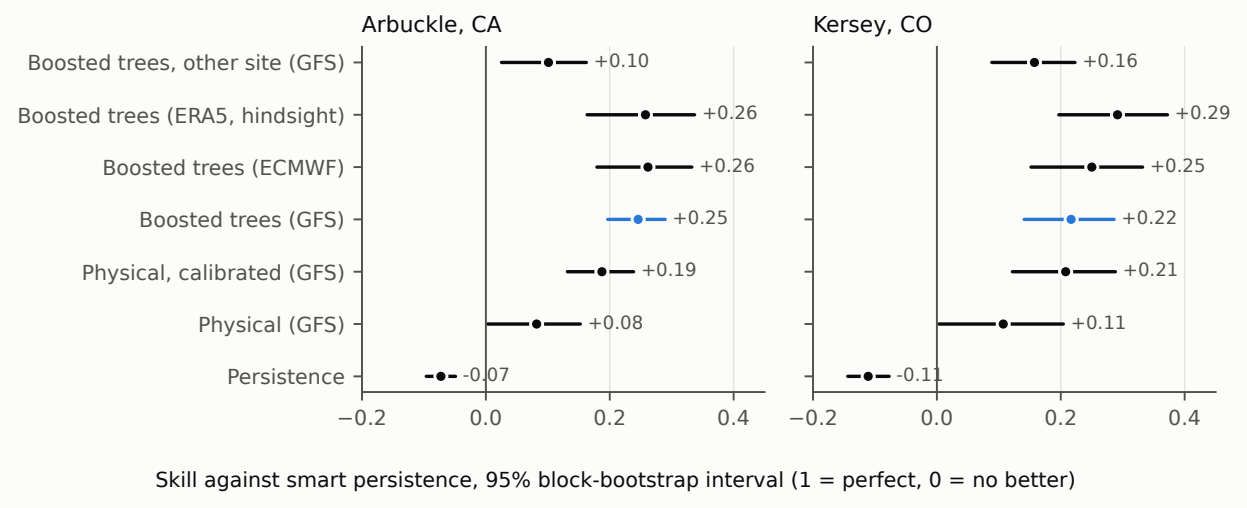


Figure 1: Skill against smart persistence with 95% block-bootstrap intervals. The primary model is highlighted.

5.2 Which weather

Replacing GFS with the ECMWF forecast changes little at Arbuckle (trees +0.26) and helps at Kersey (+0.25); the uncalibrated physical model gains more from ECMWF at both sites (+0.18 and +0.24), which suggests the GFS irradiance carries a bias the calibration and the trees mostly learn away. Giving the trees the weather that actually happened (ERA5) as if it had been forecast lifts skill only to +0.26 at Arbuckle and +0.29 at Kersey. The day-ahead weather forecast is therefore not the main limit on these scores. At Arbuckle in particular, replacing the forecast with the reanalysis adds almost nothing; most of the remaining error is in how weather becomes power at this plant, in local conditions the reanalysis does not resolve either, and in the plant itself.

5.3 Sky condition and season

Figure 2 splits the errors by the day's measured clearness. On clear days, which are 343 of 549 at Arbuckle and 161 of 298 at Kersey, the trees' error is lowest (nRMSE 0.103 and 0.142) and their skill highest at Arbuckle (+0.40), likely because smart persistence is wrong on every clear day that follows a cloudy one and a weather forecast need not be. On overcast days errors are largest for every forecaster (trees 0.248

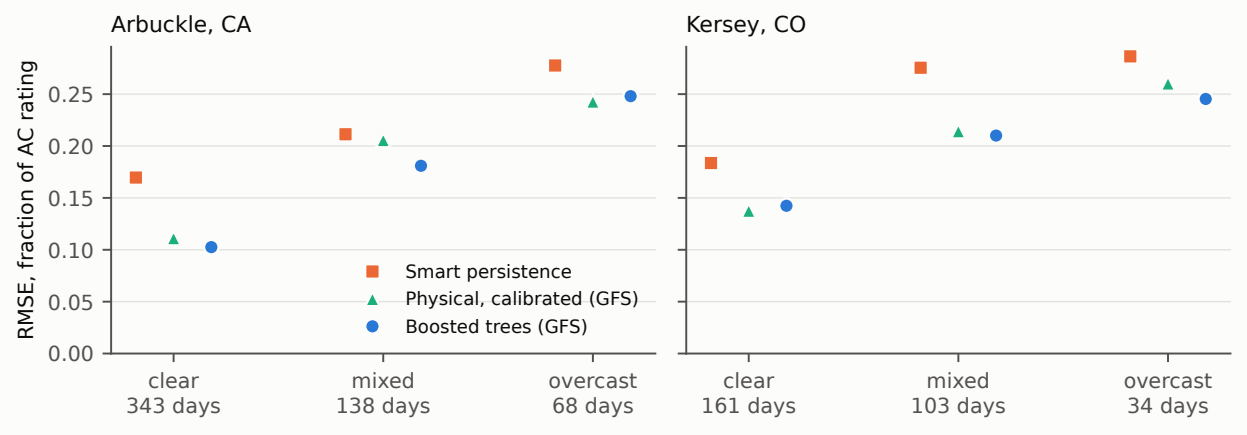


Figure 2: RMSE by the day's measured clearness (energy over clear-sky energy). Clear ≥ 0.8, overcast < 0.4. The classes use the outcome and describe where error falls, not a forecastable state.

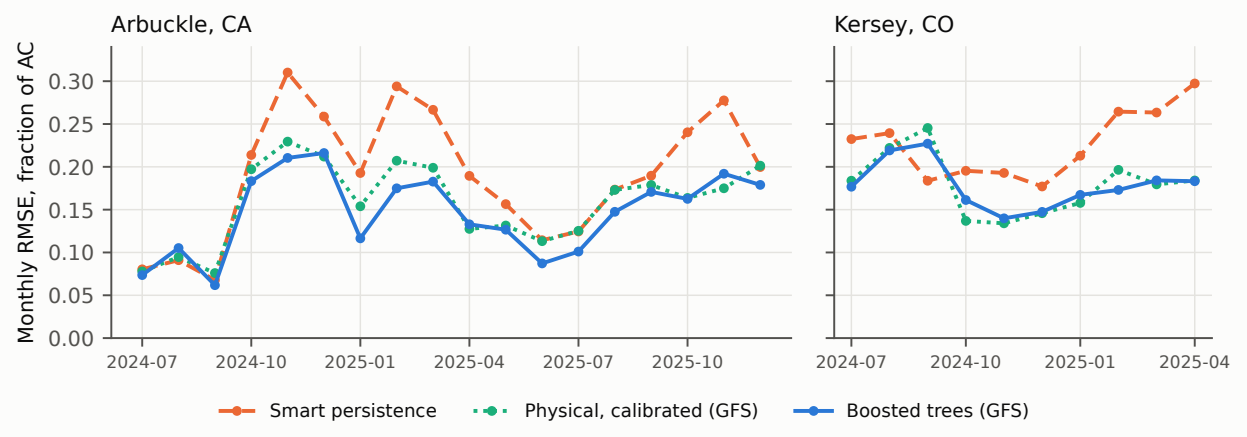


Figure 3: Monthly RMSE for smart persistence, the calibrated physical model and the boosted trees (GFS inputs).

and 0.245) and skill is lowest (+0.11 and +0.14). At Kersey the calibrated physical model is as good as the trees on clear days (+0.25 against +0.22); the trees' small advantage there is on mixed and overcast days.

By season (Figure 3), Arbuckle's summers are easy for every forecaster: in June to August smart persistence scores 0.121 and the trees 0.105, a skill of only +0.13. Winter and spring are where forecasting pays: in December to February the same numbers are 0.240 and 0.174, a skill of +0.27, and spring is similar (+0.28). Kersey's seasons are single instances, and two are partial (summer is July and August 2024, spring March and April 2025). Its summer is weaker than its winter (+0.16 against +0.26), its weakest season is autumn (+0.05) and its strongest spring (+0.35); its smart persistence gets steadily worse through the winter and spring of 2025 while the weather-driven forecasts hold steady. Arbuckle's last months of 2025 stand out and are taken up in Section 6.

5.4 Transfer between sites

A model trained only on the other plant, then given this plant's inputs, keeps a positive skill at both sites but loses much of it: +0.10 (+0.03 to +0.16) at Arbuckle and +0.16 (+0.09 to +0.22) at Kersey, against +0.25 and +0.22 when trained at home. The difference in RMSE, home model minus transferred model, has a 95% interval of −0.036 to −0.020 of rating at Arbuckle and −0.025 to −0.005 at Kersey. Both models see physical

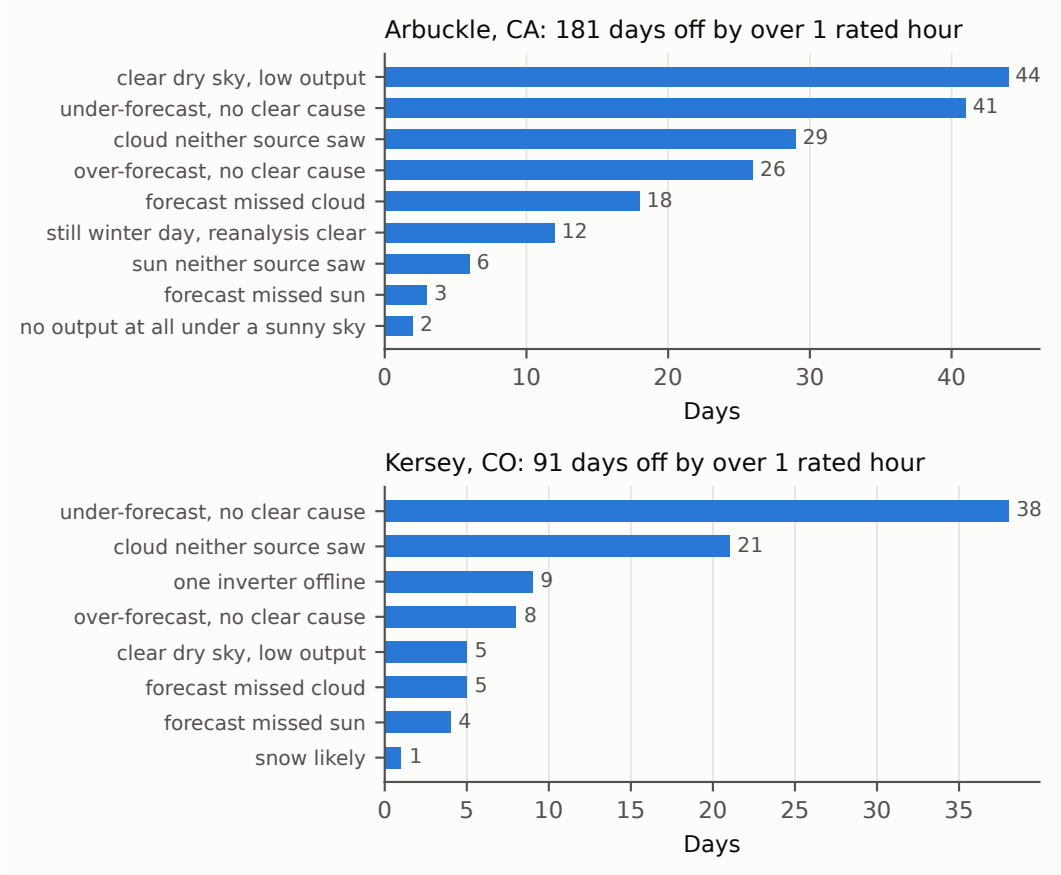


Figure 4: Days with a daily energy error above one AC-rated hour, sorted by the fixed rules in `pipeline/failures.py`.

and clear-sky power in units of each plant's own rating, so geometry and size are largely normalised away; what does not transfer is the relation between this site's weather forecasts and this plant's behaviour. That relation is, in a direct sense, "about one place". At both sites the transferred model's skill falls between that of the uncalibrated physical model (+0.08 and +0.11) and the calibrated one.

5.5 Intervals

The 10–90% quantile band should hold 80% of hours. It holds 59% at Arbuckle and 65% at Kersey (mean width 0.209 and 0.302 of rating; mean pinball loss 0.0310 and 0.0390). The bands are too narrow at both sites, and hindsight weather does not fix it (ERA5: 54% and 64%). A likely reason is that quantile trees trained on a year or less learn the spread of errors in their training months, while the largest errors in the test months come from plant faults (described next) that no training month resembled.

6 Failure analysis

Error is concentrated: the worst 10% of days carry 25% of all absolute error at Arbuckle and 24% at Kersey (46% and 37% of squared error). Of the scored days, 181 at Arbuckle and 91 at Kersey were off by more than one rated hour of energy. Figure 4 sorts those days by the rules above.

The plant, not the sky. The single largest miss at Arbuckle is 18 October 2024: a clear day by the forecast and the reanalysis, and zero output all day (Figure 5, top left). 6 November 2024 is the same. The data cannot say whether the plant was off or only its meter; either way, no weather forecast could have known. At Kersey, inverter 2 reported zero for days in mid-September 2024 while inverter 1's output was capped flat (top right); those days are five of Kersey's fifteen worst. From March 2025 Kersey's inverter 1 ran at a median power factor of 0.86 while producing, against 1.00 before, consistent with a reactive-power setting. It does not explain the misses: on days the reanalysis called clear, the plant's median clearness was 0.93 from March (34 days) against 0.96 before (159 days), a difference the change of season alone could produce. The power factor falls further on cloudy days, when output is low for other reasons, which is why the earlier rule that read a low power factor on a low-output day as a cause was dropped. At Arbuckle, 44 days were clear and dry by both weather sources and the plant still made far less than clear-sky output (mean k_{obs} 0.64 against k_{era5} 1.06); almost all fall between May and September 2025. Equipment, soiling and wildfire smoke would all look like this in the data used here, and they cannot be separated without on-site measurements.

Still winter days. 12 Arbuckle days in November to February had light wind, a reanalysis sky close to clear (mean k_{era5} 1.03), a day-ahead GFS forecast of partial cloud (mean k_{nwp} 0.69) and a plant that saw far less light than either (mean k_{obs} 0.21). All of them fall in November and December 2025 (Figure 5, bottom left). This pattern is consistent with radiation fog in the Sacramento Valley, a known winter phenomenon there [3], which a coarse model can miss. The data here do not prove it was fog. On these days the day-ahead forecast was closer to what happened than the reanalysis was.

Weather. Only a minority of the large misses are a weather forecast that was wrong in a way the reanalysis later corrected: 18 days of missed cloud and 3 of missed sun at Arbuckle, 5 and 4 at Kersey. More common are days when both weather sources agreed and the plant saw something else: cloud that neither source saw (29 and 21 days). Kersey's 8 to 10 August 2024 (Figure 5, bottom right) shows three kinds of miss in three days. On the 8th both weather sources called for heavy cloud and the trees still forecast several times the energy the plant made, so the error there is the model's. On the 9th the GFS forecast was itself much brighter than what the plant saw, and the trees forecast brighter still. On the 10th the plant saw cloud that neither source had. The largest single category at Kersey is under-forecasts with no clear cause (38 days): days that every source called clear (mean k_{obs} 0.97, k_{nwp} 0.96, k_{era5} 0.97) on which the trees still predicted too little. Their cause was not investigated.

How much the plant matters. Removing the plant-anomaly days (59 at Arbuckle, 21 at Kersey; the rule uses no forecast) raises the trees' skill from +0.25 to +0.29 (+0.24 to +0.34) at Arbuckle and from +0.22 to +0.25 (+0.19 to +0.31) at Kersey. A forecast evaluated on a clean plant looks better than one evaluated on a real one, and the difference is comparable to what the trees add over the calibrated physical model at Arbuckle.

7 Limitations

- Two plants. Neither is a cloudy humid site; the contrast is a Mediterranean valley against a semi-arid plain, because those were the plants with public data in the forecast archive's period. Kersey has 16 months, ten of them scored.
- Short training records. The first test month has about five and a half months of history. Learned models, and especially the quantile bands, would likely improve with more years; this study cannot say by how much.

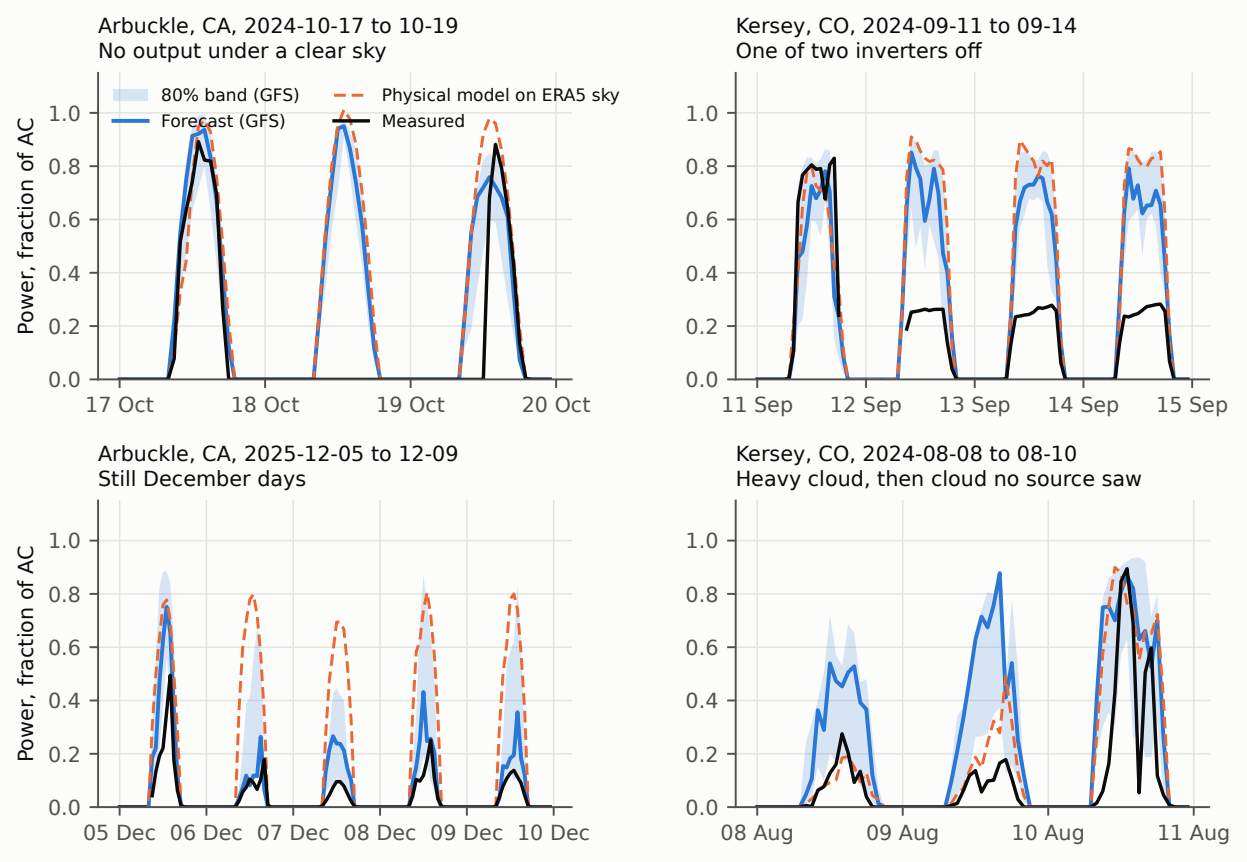


Figure 5: Four failure cases: measured power, the GFS-driven trees with their 80% band, and the physical model on reanalysis (ERA5) weather.

- Hourly, single-point weather. Inputs are the hourly values at the nearest grid point. No spatial or ensemble information is used.
- The plant faults are inferred from power, power factor and weather alone. Without maintenance logs, the categories "no output", "clear dry sky, low output" and "still winter day" are descriptions, not diagnoses.
- Fixed, untuned hyperparameters. This avoids optimism, but a tuned model might do somewhat better.
- The Open-Meteo archive is a third-party reconstruction of past runs. The pipeline pins dates and records a checksum of every response, but cannot rule out later changes to the archive.

8 Reproducibility

Everything runs from one command, `python -m pipeline all`: it downloads the raw files (about 30 MB) and stops if any differs from the committed SHA-256 manifest, cleans the series and writes the ledger, builds the features, runs the rolling-origin evaluation, sorts the failures, draws the figures and writes the macros this paper's numbers are typeset from. A full run takes a few minutes on a laptop. Seeds are fixed and LightGBM runs single-threaded in deterministic mode; a rerun from an empty data folder reproduced every committed result file exactly. Unit tests cover the cleaning rules and metrics, and an end-to-end

test on a synthetic plant checks that the pipeline ranks hindsight weather above the forecast and both above smart persistence and that a planted leak is caught. The code is MIT-licensed at https://github.com/speyyman/solar-two-sites; this text and its figures are CC BY 4.0. Data sources and licences are listed in the repository's `DATA.md`.

Declarations

Data availability. All data are public. Plant measurements are from the NREL PVDAQ archive on the Open Energy Data Initiative (systems 2107 and 9068, CC BY 4.0). Day-ahead forecasts and reanalysis are from Open-Meteo (CC BY 4.0). The exact queries and a SHA-256 checksum for every downloaded file are recorded in the repository's `data/manifest.json`; sources and licences are listed in `DATA.md`.

Code availability. The full pipeline, from the raw downloads to every number, table and figure in this paper, is at https://github.com/speyyman/solar-two-sites under the MIT licence, and is archived with this paper at doi:10.5281/zenodo.23604923.

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A Cleaning ledger

Every value the pipeline dropped, shifted or filled in, by rule. “Power” rows are the measured output used for every score; “POA” rows are the on-site irradiance sensors, which were checked but not used; “features” rows are fills in the persistence inputs.

Site	Stage	Rule	Count	Unit
Arbuckle	power	timestamp ambiguous or nonexistent in local clock; dropped	12	samples
Arbuckle	power	overlapping timestamps across files; first file kept	29,175	samples
Arbuckle	power	clock shifted to match solar noon	70,159	samples
Arbuckle	power	negative power (night tare) set to 0	0	samples
Arbuckle	power	above 105% of maximum AC output; dropped	0	samples
Arbuckle	power	identical non-zero value for 60 min or more; dropped	153	samples
Arbuckle	power	hour with under 75% of its samples; set missing	32	hours
Arbuckle	POA	clock shifted to match solar noon	178,864	samples
Arbuckle	POA	month whose clock disagrees with the sun by more than 0.75 h; dropped	6,736	samples
Arbuckle	POA	overlapping timestamps across files; first file kept	52,524	samples
Arbuckle	POA	hour with under 75% of its samples; set missing	10,973	hours
Kersey	power	sample missing in source	1,458	samples
Kersey	power	timestamp ambiguous or nonexistent in local clock; dropped	12	samples
Kersey	power	negative power (night tare) set to 0	1,561	samples
Kersey	power	above 105% of maximum AC output; dropped	0	samples
Kersey	power	identical non-zero value for 60 min or more; dropped	0	samples
Kersey	power	hour with under 75% of its samples; set missing	52	hours
Kersey	POA	timestamp ambiguous or nonexistent in local clock; dropped	7	samples
Kersey	POA	clock shifted to match solar noon	137,153	samples
Kersey	POA	month whose clock disagrees with the sun by more than 0.75 h; dropped	8,791	samples
Kersey	POA	hour with under 75% of its samples; set missing	32	hours
Arbuckle	features	smart persistence: D-1 incomplete, ratio taken from an older day	31	days
Arbuckle	features	persistence: D-1 missing, filled from an older day (up to 7)	52	day-hours
Kersey	features	smart persistence: D-1 incomplete, ratio taken from an older day	36	days
Kersey	features	persistence: D-1 missing, filled from an older day (up to 7)	351	day-hours